



OHIO RIVER TRAIL NORTH SHORE FEASIBILITY STUDY

December 5, 2011 Public Meeting

PRELIMINARY REVIEW OF ROUTING ALTERNATIVES OPTIONS AND FINDINGS

Section 100

Constraints

1. The portion of Midland Road (Route 68) entering Ohio is heavily trafficked with a speed limit of 45 mph and not conducive to a shared bicycle path. Steep embankments on either side of the existing road make this situation more problematic. Rockslides could leave debris on the roadway that would be harmful to both motorists and cyclists.
2. Smiths Ferry Road provides ramped access from Glasgow Borough, underneath the railroad and Route 68, then wraps around and onto Route 68. However, as the shared bicycle lane would require bicycle travel in both directions, eastbound traffic would need to cross Route 68 at grade to continue towards Midland.
3. The varying topography along this portion of Route 68 makes the prospect of split bicycle and vehicular paths challenging. Route 68 and its paralleling railroad are cut into a hillside that slopes steeply down to the Ohio River.

Options/Alternatives

1. Alternative A – Glasgow Borough could be used as an alternate trail route to Route 68. The trail could diverge from Route 68 along Main Street in Glasgow, travel under the train tracks via an existing tunnel, and then continue east along Liberty Avenue to Smiths Ferry Road.
Alternative B – Extending the trail into Ohio via Tuscarawas Road just north of the Main Street tunnel in Glasgow could help to remove bicycle traffic from Route 68 and could provide a direct linkage to the Great Ohio Lake Greenway which begins just over the state line in Ohio.
2. A street crossing with flashing signage and on road painting could help make the crossing from Smiths Ferry Road to Route 68 eastbound a safe and viable option for cyclists.
3. There is another ledge cut into the topography above Route 68 away from the Ohio River. However, this ledge, which could become a dedicated bicycle path, travels through several private parcels of land between Smiths Ferry Road and the Ohioville/Midland border. While this ledge presents a more desirable trail alternative, it could be difficult to achieve due to land ownership.

Section 200

Constraints

1. Route 68 entering Midland from the west is a narrow stretch of road between 20' and 24' wide with varying and at times nonexistent shoulders isolated by steep topography in both directions.
2. In Midland, the Ohio River bends away from existing thoroughfares. The land between the towns main road (Route 68) and the Ohio River is privately owned and is still being used for industrial purposes. Though paralleling the Ohio River as much as possible is at the core of this projects vision, achieving this design in Midland could prove challenging.
3. In Midland, following the Ohio River closely could jeopardize potential revenue generation for the downtown. This potentially missed opportunity is due to the fact that the Ohio River is separated from the downtown portion of Midland by several large, operating, industrial sites.
4. Railroad Avenue, which is being considered as an alternate trail alignment for this section of the Ohio River Trail, is a one-way street eastbound.
5. Shipping Port Bridge connecting Midland to its industrial sites across the Ohio River presents some circulation challenges. Traffic speeds increase here as vehicles leave downtown Midland. This portion of Route 68 is also heavily traveled by trucks from various industrial sites in Midland heading east.

Options/Alternatives

1. While the uphill or north side of Route 68 into Midland presents construction complications the south side or Ohio River side could be expanded into a two-way bicycle path. However, this is not the ideal situation as it would then lead to a greater number of road crossings for bicyclists.
2. Having the trail parallel the Ohio River could cause unwanted interference with industry still active in Midland. This could also lead to missed economic opportunities for Midland Borough. Therefore, one alternate trail alignment would be along Railroad Avenue just north of the industrial sites and south of downtown Midland. The low traffic, 25 mph speed limits and long sightlines make this alternate an attractive option.
3. The Railroad Avenue alternate path is close enough to downtown to foster economic growth in downtown Midland. Ideally, cyclists and pedestrians using the trail would stop into town for refreshments and services. Being only one block south of the main thoroughfare, Railroad Avenue makes this economic spillover possible.
4. Railroad Avenue being a one-way street makes the idea of a split eastbound / west -bound bicycle route an appealing one. The westbound route could occur along Beaver Avenue, one block north of Midland Avenue. Utilizing Beaver Avenue or one of the other northern avenues would also help to connect the Ohio River Trail to Midland's cultural assets such as the historic Carnegie Library and the adjacent Lincoln Park. These northern avenue options are also appealing because of their existing characteristics. Their historic brick paving patterns help to calm vehicular traffic. This calming is

further enforced by a 25 mph speed limit. That being said, brick paving is not ideal for bicycle traffic and portions may have to be paved for the bicycle trail.

5. The trail could follow existing vehicular traffic patterns around Shipping Port Bridge, splitting eastbound and westbound lanes. The eastbound portion of the trail could go under the bridge while the westbound portion could utilize the 7' shoulder abutting the hillside to the north.

Section 300

Constraints

1. There is a pinch point along Route 68 just east of Wabash Road heading towards Industry. Route 68 becomes 22' wide with a 2' paved shoulder on north side and an 8' unpaved shoulder on the south.
2. Beyond this pinch point heading east Route 68 becomes 24' wide with 5-6' paved shoulders. The shoulders are not ideal for bicycle traffic as they currently exist.
3. To the east of Barclay Hill Road Route 68 bridges the railroad. Though the bridge has wide, designated shoulders, it is curved reducing sight distances and not currently conducive to pedestrian traffic.

Options/Alternatives

1. Alternate A – To avoid this pinch point the trail could be directed away from Route 68 along Wabash Road. Heading south along Wabash Road the trail could then take Canton Street, a low traffic neighborhood street, until it rejoins with Route 68 past the pinch point.

Alternate B – The trail could travel north along Wabash Road and link with Pleasant Street eastbound, then to Industry Drive and Terrace Lane. This alternate option becomes the more viable of the two if the trail is to continue to run to the north of Route 68 along Six Mile Run, into Industry.

2. Route 68 through Industry could be redesigned to be more bicycle safe. Designating the shoulders as dedicated bicycle lanes with paint markings and rubber strips would help to make the most of this linear section of Route 68 that already provides great sight distances.
3. Alternative A – The bridge over the railroad could be adapted to support bicycle traffic. The shoulders could be protected from the road by guard rails and then turned into dedicated bicycle lanes. There would also need to be some sort of fencing protecting pedestrians from a fall over the edge of the bridge to the railroad below.

Alternative B – The trail could go under the vehicular bridge and cross the railroad at grade. This would raise other issues such as adding an at grade crossing but should still be considered an alternative to the vehicular bridge crossing.

Alternative C – The trail could stay to the north of Route 68 and avoid crossing the railroad all together. This alternative would mean the construction of a path solely dedicated to bicycles and would require more planning than a shared path along Route 68 would. However, it would also solve another constraint further east along Route 68 where the road crosses back over the railroad, again via a bridge.

Section 400

Constraints

1. There is a pinch point approximately 100' east of the Route 68/railroad bridge mentioned in the Constraints portion of Section 300. Here, Route 68 narrows to 23' with a 5' shoulder on the north side and a 3' shoulder with a guardrail on the south. Both shoulders are separated from the traffic lanes by rumble strips.
2. One of the narrowest portions of Route 68 occurs west of Vanport near the Lockhouse 6 Restaurant. The road becomes 23' wide with 1' shoulders on either side. The shoulders are framed by curbs, a slope to the north and a series of utility lines to the south.
3. East of the Lockhouse 6 Restaurant there is another pinch point. Here, in front of Interstate Chemical, Route 68 is 23' wide with a 1' shoulder and sloped hill to the north and a 1' shoulder and 3' span to a chain linked fence on the south. This situation would not leave much room for bicycles if this were to become a shared portion of the path.

Options/Alternatives

1. Halfway between the two Route 68/railroad bridge crossings there is a tunnel to the north that passes beneath the existing railroad. This tunnel could be used as a safe and unobtrusive bicycle crossing point. Other issues would then need to be addressed such as how the trail would continue north of the railroad through what is largely private property.
2. This section of Route 68 could be avoided by running the trail to the north of the railroad from the existing tunnel into Vanport. The trail could then rejoin the existing street network at Sebring Road.
3. Alternative A – This second pinch point to the east of Lockhouse 6 Restaurant could be avoided by running the trail to the north of Route 68 then eventually onto Sebring Road.

Alternative B – Although currently constrained by various features such as the slope to the north and the chain linked fence to the south, these features are not necessarily permanent. It may be possible, through various negotiations, to move the fence further south allowing for the shoulders on either side to open wider. Currently they do expand closer to the Lockhouse 6 Restaurant to 4' on the north and 10' on the south.

Section 500-A

Constraints

1. Bringing the North Shore Trail into Vanport via Division Lane is ideal as it avoids the traffic that occurs when Route 68 intersects with the Beaver Valley Expressway (Interstate 376). However, traveling along Division Lane moves the trail farther north away from the Ohio River.
2. Buffalo Road resting on the Vanport/Beaver line and running north/south is a thoroughfare of relative traffic that would need to be crossed in order for the North Shore Trail to reach central Beaver.
3. In Beaver there exists a strong network of low traffic neighborhood streets ideal for cyclists. These streets are laid out in a rectilinear fashion roughly following cardinal directions, making movement from east to west and north to south possible. The constraint in this situation is that certain streets are one-way.
4. Making the connection between Beaver and Rochester over the Beaver River bicycle safe is a pivotal aspect of this project. Without it, a direct connection to the proposed Ohio River South trail would be lost. That being said, this connection is not easily made within the framework of the existing infrastructure.
5. Having a North Shore spawn trail run north along the Beaver River through the Rochester River Front Park would be ideal for connecting to communities such as New Brighton, Beaver Falls and Patterson. This portion of the trail would need to cross McKinley Run which currently is not possible for bicyclists.

Options/Alternatives

1. Alternative A – To bring the trail south again from Division Lane it could travel south along Spring Lane around the Vanport Ballfields, then connect with Georgetown Lane north of the Beaver Cemetery. Portions of this segment could be shared paths along roads with light vehicular traffic. The portion north of the cemetery could be a dedicated bicycle path separated from Georgetown Lane by an existing swale. Site observations have made it apparent that what would become a dedicated bicycle path adjacent to the cemetery is already in use as a footpath connecting the Vanport Ballfields with other community resources such the Beaver County High School.

Alternative B – Another option would be to have the trail enter as it does in Alternative A, along Division Lane, then travel south along Spring Lane and continue weaving its way southeast via Route 68 and Sassafras Alley, eventually connecting with River Road which runs east/west paralleling the Ohio River. However, this creates a similar situation as to the one described in Midland where the trail would be directed away from the main commercial center of the town in order to parallel the Ohio River. As described in the Midland situation, this could result in the potential loss of economic gains for Beaver.

Alternative C – The trail could enter Vanport along Route 68 (which becomes 3rd Street in Beaver). This would help to keep the trail closer to the Ohio River. However, this alternative would also mean

that the intersection of Route 68 with Interstate 376 would need to be reengineered. Reengineering this section presents its own issues as it is already a complicated network consisting of a bridge with several on and off ramps.

2. Assuming that the North Shore Trail enters Beaver north of the Beaver Cemetery, a shared path along Tuscarawas Road becomes an appealing option for crossing Buffalo Road. Currently, where Tuscarawas Road intersects with Buffalo Road there is a traffic light. Keeping this traffic light and adding a signal for cyclist sharing Tuscarawas Road with motorist makes the connection into downtown Beaver possible and safe.
3. Alternative A – The one-way street network in Beaver is conducive to one-way bicycle paths. The Ohio River Trail could become a series of shared road paths working with the existing circulation directions. Canal Street could carry cyclists westbound as could Corporation.
Alternative B – This issue could be avoided by keeping the trail as a shared path from Tuscarawas Road to 4th Street which is a two-way street. 4th Street is also a 40' wide street with 6' sidewalks on either side. These existing dimensions make adding bicycle lanes possible.
4. Alternative A – The new Veterans Memorial Bridge is under construction along the northern section of the Beaver River near New Brighton. This new bridge will have a shared bicycle/pedestrian path along its southern edge. In terms of connecting Beaver with Rochester this is not the ideal solution because it is too far north to be considered convenient.
Alternative B – Latching onto or creating a shared path along an existing Beaver/Rochester bridge is worth considering. It would provide direct crossing and could do so without a great deal of additional expense. While this may be possible for the Rochester/Beaver Bridge and the Rochester/Bridgewater Bridge, doing this along the existing railroad bridge is questionable. Using such an active rail bridge as the backbone of an attached bicycle bridge does not create the most enjoyable bicycling experience, though it is the closest existing bridge to the Ohio River.
Alternative C – Building a pedestrian/bicycle bridge spanning the Beaver River, though perhaps ideal, would be costly. One alternative to a spanning bridge would be to create some sort of pontoon or floating bridge that would rest upon the Beaver River. The issue then becomes boat traffic moving from the Ohio River to the Beaver River.
5. A safe crossing for McKinley Run could be resolved with a prefabricated bridge. Such a bridge could facilitate pedestrian and bicycle traffic.

Section 500-B

Constraints

1. Along the eastern shore of the Beaver River is the Rochester River Front Park. This park has the potential to serve as a bicycle path linking the Rochester portions of the North Shore Trail to other communities further north along the Beaver River such as New Brighton, Beaver Falls and Patterson. This section of trail is constrained by the topographic conditions that exist between the river and the surrounding towns.
2. Crossing the Beaver River to connect New Brighton and Beaver Falls would also connect the proposed Ohio River Trail to an existing bicycle trail that begins near Morrell Field. However, 7th Street Bridge is the only existing pedestrian crossing point.
3. While Beaver Falls has many existing streets that could support a shared bicycle path system, choosing the right one is important. Many of the streets have wide traffic lanes and wide sidewalks but are less than ideal due to poor lighting and surrounding context.
4. A connection between the proposed Ohio River Trail and Bradys Run Park is important as the park is one of the region's largest natural resources and a place visited often by hikers and cyclists. To make that connection under current conditions involves a dangerous journey along Constitution Boulevard (Route 51). Route 51 has very narrow shoulders (nonexistent in some places), a high volume and speed of vehicular traffic and limited sight lines.

Options/Alternatives

1. There are ramping options that would allow for a riverside trail to overcome the riverbank topography and meet with existing town roads such as YMCA drive in Rochester. One of these ramping options would also connect a riverside trail with the future Veterans Memorial Bridge, currently under construction.
2. The existing 7th Street Bridge that connects New Brighton and Beaver Falls has a pedestrian crossing separated from traffic by a concrete guardrail. While this provides safe crossing once on the bridge, getting onto and off of the bridge remains an issue for cyclists. To resolve this issue in New Brighton, a bicycle trail could be wrapped along 4th Street and then down 2nd Avenue. From 2nd Avenue, the trail could connect with Big Rock Park and Fishing Park.
3. 6th Avenue seems to be one of the best options for a shared bicycle path through Beaver Falls as it is one of the main streets in the community. The trail could run along 6th avenue from the 7th Street Bridge until turning east on 11th Street. At the end of 11th Street it would be able to connect to Morrell Field and the existing bicycle system to the north. This option would also help to connect several of the smaller pocket parks found in Beaver Falls.
4. Alternatives are still being considered to bring the proposed Ohio River North Trail into Bradys Run Park. One option is to use another road as a shared bicycle path avoiding any cycling along Route 51.